

Satellite Building Workshop



Space Faculty Private Limited

At Space Faculty, we chart pathways between earthly dreams to interstellar reality. We power brilliant and bold ideas in Space and frontier technologies through training, engineering, research and strategic public-private partnerships.

www.spacefaculty.asia



About Space Faculty Pte Ltd

Our expertise spans several key areas:

- 1. Satellite Development:** From satellite building and launching to providing hands-on training in satellite development.
- 2. Scientific Advancement:** Facilitating cutting-edge space experiments for the scientific community.
- 3. Workforce Development and Executive Programmes:** Specialized Executive courses, training, workshops and hackathons to understand how space technologies can future proof your organization and your career
- 4. National-level Space and STEM 2.0 Roadmap Development:** Executing impactful, roadmaps through training, "galaxy-class" competitions and hackathons with global experts to ignite youth interest in space and STEM fields.

HONORARY ADVISORS

- Prof Lui Pao Chuen, Mr Atsushi Taira, Mrs Koh Soo Boon, Mr Seah Moon Ming, Dr Victoria Coleman

CEO

- Ms Lynette Tan

Ang Eng Hian

- Education:
 - NUS BEng (EE)
 - NUS MSc (EE)
 - NUS MTech (SE)
- Space Faculty
 - Technical Consultant
- DSO National Laboratories
 - 2007 to 2025
 - Principal Engineer (Radar System Engineer, Satellite Payload System Engineer)
 - RF system, Electronic, Software, FPGA, Signal Processing

Nelson Tan

- Education:
 - NTU BEng (ME)
- Space Faculty
 - Satellite Engineer

SPEAKER PROFILE

Mr Nelson Tan is a Mechanical Engineer at Space Faculty, a Singapore-based SME driving innovation in the space and technology sector. His professional focus centers on spacecraft structural design, precision manufacturing, and system integration. Before joining Space Faculty, Nelson was a Research Engineer at Thales Solutions Asia and Nanyang Technological University's Satellite Research Centre (NTU SaRC). At NTU, he was instrumental in developing VELOX-AM, a collaborative mission between A*STAR's Advanced Remanufacturing and Technology Centre (ARTC) and NTU that demonstrated 3D printing technology and advanced space materials in orbit. He oversaw mechanical design, CNC fabrication, and environmental testing, ensuring compliance with ECSS/NASA launch standards. Nelson applies his multidisciplinary engineering background to transform complex technical concepts into practical, reliable systems. In this special

Lam Lap

- Education:
 - MSc in Electrical Engineering (UIUC, USA)
- Space Faculty
 - Technical Director
- DSO National Laboratories
 - 2003 to 2024
 - Programme Director

Agenda

S/N	Day	Time		Title	2nd Space Slides	Trainer	Duration
		Start	End				
1	1	9:00	10:00	INTRO Workshop agenda, hands-on objectives, Attendance, Final Exercise, SBJ, Team Building		AEH, E1	60
2		10:00	10:15	Satellite and Space Ecosystem		NT	15
3		10:15	10:30	Satellite & Space Mission	L1 Satellite and Space Mission	AEH	15
4		10:30	10:45	Satellite Payloads & Missions	L19 (part) Payloads and Examples of Satellite Mission	AEH	15
5		10:45	11:00	Break			15
6		11:00	11:45	Satellite Architecture, Design & Development Overview	L3 Satellite Sub-systems	AEH	45
7		11:45	12:45	Lunch			60
8		12:45	13:15	[Lab] TinkerCad with arduino(Cubesat Digital Twin)	L11 Review and Programming (Workshop Materials), New	AEH	30
9		13:15	14:45	[Lab] Cubesat hands on (inventory list chk, GS assembly)	Slides Day1 (Arduino IDE, driver setup), Day2 (GS assembly, test)	AEH, NT	90
10		14:45	15:00	Break			15
11		15:00	17:00	[Lab] Cubesat hands on (Cubesat assembly)	Slides Day3 (cubesat assembly)	AEH, NT	120
12	2	9:00	10:15	Command & Data Handling Subsystem (CDHS)	L6 Satellite Command and Data Handling	AEH	75
13		10:15	10:30	Break			15
14		10:30	11:00	Thermal Control Subsystem (TCS)	L9 Satellite auxiliary systems_Thermal_Radiation	AEH	30
15		11:00	11:30	Electrical Power Subsystem (EPS)	L4 Electrical Power System	AEH	30
16		11:30	12:30	Lunch			60
17		12:30	13:45	[SI] AIT, Project Management, mission management	Slides by Space Inventor, L18 Mission management development, L10 S	Nuno	75
18		13:45	15:15	[Lab] Cubesat hands on (assembly, programming, initial tests)	Slides Day3 (cubesat assembly), Day4 (OBC)	AEH, NT	90
19		15:15	15:45	Break			30
20		15:45	17:15	[Lab] Cubesat hands on (TM, LoRa)	Slides Day4 (OBC Temperature, LoRa)	AEH	90

Agenda

S/N	Day	Time		Title	2nd Space Slides	Trainer	Duration
		Start	End				
20	3	9:00	10:30	Orbital mechanics	Naga's Slides	NT/Naga	90
21		10:30	11:00	[Lab] Space Mission Design (GMAT)	L14 Space Mission Design SW [GMAT]	AEH, NT	30
22		11:00	11:15	Break			15
23		11:15	12:15	Orbital mechanics	Naga's Slides	NT/Naga	60
24		12:15	13:15	Lunch			60
25		13:15	15:15	Attitude Determination & Control Subsystem (ADCS)	L8 Attitude Determination and Control System or Naga's	AEH, NT/N	120
26		15:15	15:45	Break			30
27		15:45	17:15	[Lab] Cubesat hands on (Solar Panel)	Slides Day4 (EPS temperature, battery voltage TMs)	AEH, NT	90
28	4	9:00	10:30	Telemetry, Tracking and Command Subsystem (TT&C)	L7 Satellite Telecommunication System	AEH	90
29		10:30	11:00	Break			30
30		11:00	11:45	Structures & cable harness	L5 Satellite Structures	NT	45
31		11:45	12:30	Propulsion Control Subsystem (PCS)	L12 Satellite propulsion	NT	45
32		12:30	13:30	Lunch			60
33		13:30	15:00	[Lab] Cubesat hands on (TT&C)	Slides Day5 (VBatt, Solar Panel, TM real-time plotter)	AEH, NT	90
34		15:00	15:30	Break			30
35		15:30	17:00	[Lab] Cubesat hands on (camera)	Slides Day5 (camera)	AEH, NT	90

Agenda

S/N	Day	Time		Title	2nd Space Slides	Trainer	Duration
		Start	End				
36	5	9:00	10:30	[Lab] Cubesat hands on (Reaction wheel test)	Slides Day6 (reaction wheel)	AEH, NT	90
37		10:30	10:45	Break			15
38		10:45	11:15	[Lab] Cubesat hands on (Reaction wheel test)	Slides Day6 (reaction wheel)	AEH, NT	30
39		11:15	11:45	Briefing on Project Report on Mission Design	L20 Mission Design – lets design a satellite mission, L18 Mission manag	NT	30
40		11:45	12:00	[Lab] Cubesat hands on (Mission)		NT	15
41		12:00	13:00	Lunch			60
42		13:00	15:00	SaRC Visit		LWS	120
43		15:00	15:15	Break			15
44		15:15	16:45	Industrial Guest Of Honor QnA and Certification presentation		AEH	90
45		16:45	17:15	Close up of Mission Design and Cubesat Disassembly	Survey, SB Proj		30

Course Materials

<https://github.com/sbwspacefaculty-png/SatBuildWorkshop>



Notes

1. Handling of soldered cables, solar panels
2. Explain the circuit of RBF
3. Explain the need to be careful in flashing correct code
4. No sticker/tape on solar panels
5. Follow the Arduino installation instruction correctly
6. Follow the assembly slides correctly

Misc

- 75% attendance for Certificate of Completion
- Final Mission Exercise Report (to submit within 2 weeks)
 - Email to :
learn@spacefaculty.asia
- Apply for participation in Satellite Building Project:
 - Email with resume:
learn@spacefaculty.asia
 - State the period of availability

Course Materials

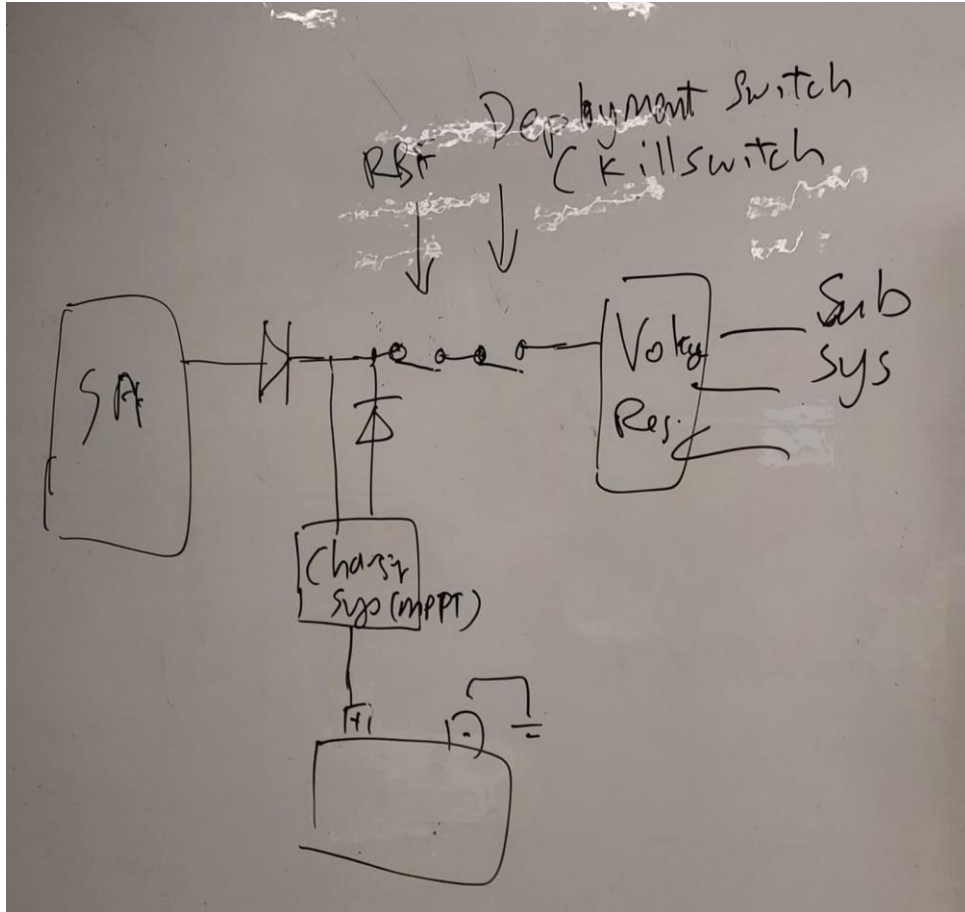
<https://github.com/sbwspacefaculty-png/SatBuildWorkshop>



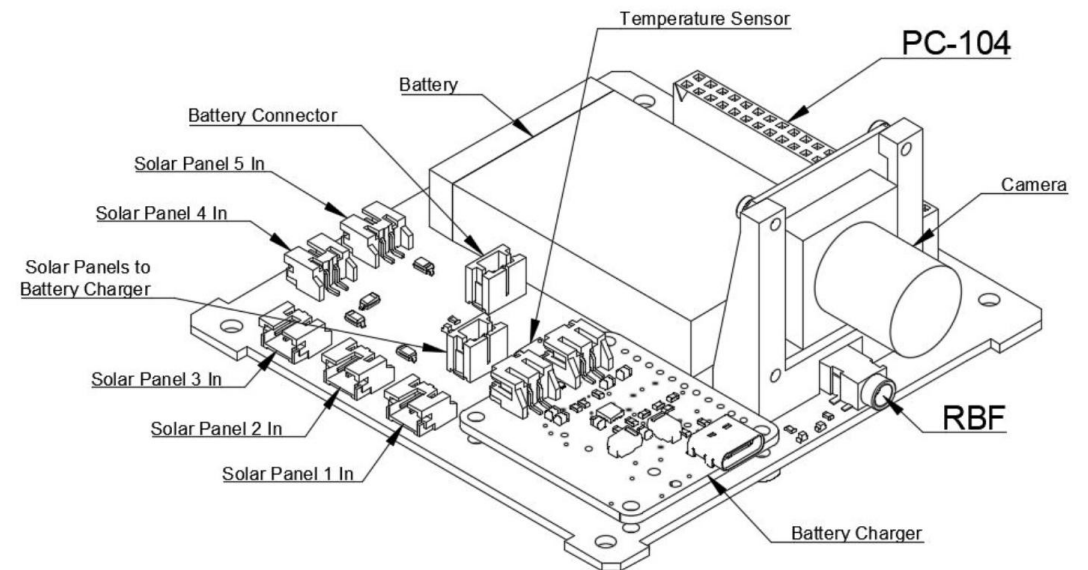
Workshop Survey



EPS Architecture and RBF



EPS Electrical Layout



Space Inventor: Nuno Andrada



horizons of invention

As an independent, Danish satellite builder rewriting the New Space playbook, we design and manufacture subsystems and platforms at our Aalborg, Denmark headquarters for optimal speed and absolute reliability.

Our unique approach is “Flexible Modularity,” using flight-proven units to create bespoke satellite platforms designed for mission-specific success.

While traditional modularity implies swapping parts within a fixed structure (a fixed box), Flexible Modularity means the structure itself is adaptable, which also speaks to our ethos as a company.



Nuno Andrada
Systems Engineer